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EXAMPLE 2.7

Given two vectors $\vec{A} = 2\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{B} = \hat{i} + 3\hat{j} + 6\hat{k}$, Find the product $\vec{A}.\vec{B}$, and the magnitudes of $\vec{A}$ and $\vec{B}$. What is the angle between them?

Solution

$$\vec{A}.\vec{B} = 2 + 12 + 30 = 44$$

Magnitude A $= \sqrt{4 + 16 + 25} = \sqrt{45}$ units

Magnitude B $= \sqrt{1 + 9 + 36} = \sqrt{46}$ units

The angle between the two vectors is given by

$$\theta = \cos^{-1}\left(\frac{\vec{A}.\vec{B}}{AB}\right)$$

$$= \cos^{-1}\left(\frac{44}{\sqrt{45 \times 46}}\right) = \cos^{-1}\left(\frac{44}{45.49}\right)$$

$$= \cos^{-1}(0.967)$$

$$\therefore \theta \cong 15^\circ$$

EXAMPLE 2.8

Check whether the following vectors are orthogonal.

i) $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = 4\hat{i} - 5\hat{j}$

ii) $\vec{C} = 5\hat{i} + 2\hat{j}$ and $\vec{D} = 2\hat{i} - 5\hat{j}$

Solution

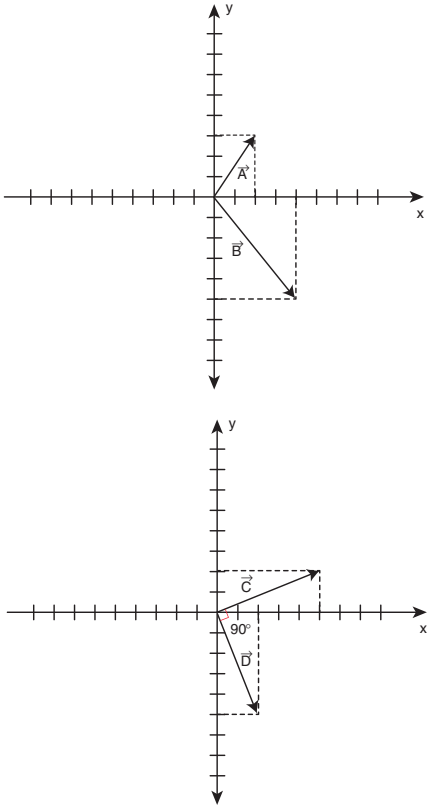
$$\vec{A}.\vec{B} = 8 - 15 = -7 \neq 0$$

Hence $\vec{A}$ and $\vec{B}$ are not orthogonal to each other.

$$\vec{C}.\vec{D} = 10 - 10 = 0$$

Hence, $\vec{C}$ and $\vec{D}$ are orthogonal to each other.

It is also possible to geometrically show that the vectors $\vec{C}$ *and* $\vec{D}$ are orthogonal to each other. This is shown in the following Figure.



In physics, the work done by a force $\vec{F}$ to move an object through a small displacement $d\vec{r}$ is defined as,

$$W = \vec{F}.d\vec{r}$$

$$W = F\ dr\ \cos\theta$$

The work done is basically a scalar product between the force vector and the displacement vector. Apart from work done, there are other physical quantities which are also defined through scalar products.

Note

In uniform circular motion, the centripetal force is perpendicular to the displacement. Hence, work done by this force is zero.